EXISTENCE OF THE INTEGRAL CLOSURE OF THE STRUCTURE SHEAF INSIDE A QUASI-COHERENT ALGEBRA ON AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a gluing argument that reduces to the scheme case on each object of the etale site; the compatibility isomorphism is what makes the construction canonical.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The integral closure of the structure sheaf inside a quasi-coherent algebra on a scheme. Morphisms, Lemma morphisms-lemma-integral-closure, used at proof lines 9946-9952. Is the scheme case applied on each U in X_etale.
- Integral closures commute with smooth (hence etale) pullback. More on Morphisms, Lemma more-morphisms-lemma-integral-closure-smooth-pullback, used at proof lines 9958-9962. Supplies the canonical isomorphism making the local data glue.
- Quasi-coherent modules on X_etale described by their values on affine etale objects. Properties of Spaces, Section spaces-properties-section-quasi-coherent, used at proof lines 9955-9957 and 9967-9972. Turns the local quasi-coherence into quasi-coherence on X.
- Restriction of a quasi-coherent sheaf along a morphism of X_etale. used at proof lines 9952-9958. Identifies A|_{U'} with the pullback of A|_U.

Source reference. Original lemma 081K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10220–10220. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 035F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4126–4126. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 03G5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4516–4516. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S. Let $\mathcal{A}$ be a quasi-coherent sheaf of $\mathcal{O}_X$-algebras. There exists a quasi-coherent sheaf of $\mathcal{O}_X$-algebras $\mathcal{A}' \subset \mathcal{A}$ such that for any affine object U of $X_{étale}$ the ring $\mathcal{A}'(U) \subset \mathcal{A}(U)$ is the integral closure of $\mathcal{O}_X(U)$ in $\mathcal{A}(U)$.*

Proof. Let U be an object of $X_{étale}$. Then U is a scheme. Denote $\mathcal{A}|_U$ the restriction to the Zariski site. Then $\mathcal{A}|_U$ is a quasi-coherent sheaf of $\mathcal{O}_U$-algebras hence we can apply Morphisms, Lemma 035F to find a quasi-coherent subalgebra $\mathcal{A}'_U \subset \mathcal{A}|_U$ such that the value of $\mathcal{A}'_U$ on any affine open $W \subset U$ is as given in the statement of the lemma. If $f : U' \to U$ is a morphism in $X_{étale}$, then $\mathcal{A}|_{U'} = f^*(\mathcal{A}|_U)$ where f^* means pullback by the morphism f in the Zariski topology; this holds because $\mathcal{A}$ is quasi-coherent (see introduction to Properties of Spaces, Section 03G5 and the references to the discussion in the chapter on descent on schemes). Since f is étale we find that More on Morphisms, Lemma 081K says that we get a canonical isomorphism $f^*(\mathcal{A}'_U) = \mathcal{A}'_{U'}$. This immediately tells us that we obtain a sub presheaf $\mathcal{A}' \subset \mathcal{A}$ of $\mathcal{O}_X$-algebras over $X_{étale}$ which is a sheaf for the Zariski topology and has the right values on affine objects. But the fact that each $\mathcal{A}'_U$ is quasi-coherent on the scheme U and that for $f : U' \to U$ étale we have $\mathcal{A}'_{U'} = f^*(\mathcal{A}'_U)$ implies that $\mathcal{A}'$ is quasi-coherent on $X_{étale}$ as well (as this is a local property and we have the references above describing quasi-coherent modules on $U_{étale}$ in exactly this manner). □